Probabilistically Robust Radar Waveform Design for Extended Target Detection

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***Abstract*—We address radar waveform design problem for extended targets under the Peak-to-Average Ratio (PAR) constraint. Based on the stochastic model of the Target Impulse Response (TIR), the distribution of output Signal-to-Interference-pulse-Noise Ratio (SINR) is derived. After that, a novel waveform design metric, named Probabilistically Robust Detection (PRD), is proposed to improve the stability of detection performance against the stochastic TIR. The geometric and limit interpretations for the PRD metric are given to show its rationality. Leveraging on the Semi-Definite Relaxation schemes and the Dinkelbach's algorithm, we devise algorithms to optimize waveform covariance matrix based on the proposed metric in high and low SINR situations, respectively. It is theoretically proved that the global optimal waveform covariance matrix can be achieved in high SINR situations. On the contrary, only the sub-optimal waveform covariance matrix can be obtained in low SINR situations. The transmit waveform is synthesized from the optimized waveform covariance matrix through randomization schemes. Numerical experiments are carried out to evaluate the PRD metric as well as the proposed algorithms. Results verify the effectiveness of the proposed algorithms. Compared with the current design metric, the proposed PRD metric exhibits strong probabilistic robustness on detecting extended targets.**

***Index Terms*—Radar waveform design, extended targets, Signal-to-Interference-pulse-Noise Ratio (SINR), Probabilistically Robust Detection (PRD) metric, Dinkelbach's algorithm.**

I. Introduction

THE concept of radar range extended target, known as extended target, was firstly proposed by Van Trees in [1], which describes the cases that the target size exceeds the radar range resolution and occupies more than one range cells in radar screen. With the expansion of radar bandwidth as well as the development of high resolution range profile techniques [2], [3], extended targets become common in practical radar detection situations. Different from the point-like targets, the returns of extended targets are not the scaled and attenuated version of transmit waveform, but the convolution between Target Impulse Response (TIR) and transmit waveform [4], [5], [6], [7]. Some conventional processing techniques for point-like targets are not the optimal choices for extended targets. For example, match filter is the optimal filter on maximizing the output Signal-to-Interference-pulse-Noise Ratio (SINR) for point-like targets. However, this is not true for extended targets. As a result, optimizing the radar transmitter and receiver for extended targets has been a hot topic during the past decades [5], [7], [8], [9], [10], [11].

A. Related Work

Waveform design techniques, improving radar detection and recognition performance by taking full advantages of transmit freedom, have drawn considerable attention from researchers and practitioner [12], [13], [14]. The seminal work of Bell was the beginning of optimizing radar waveforms for extended targets [5], where waveforms for optimal detection and information extraction are investigated. For the optimal detection waveform, maximizing the output SINR with a deterministic TIR is considered. For the information extraction waveform, the TIR is described by a random vector, and the Mutual Information (MI) between TIR and the received signal is maximized by designing transmit waveform. Besides, the minimizing Minimum Mean Square Error (MMSE) metric is also used for TIR parameter estimation or range profiling (see [8], [11], [15] for details). In [8], results show that minimizing MMSE and maximizing MI lead to the same optimal waveforms in additive white Gaussian channel.

Note that the detection probability monotonically increases with respect to the output SINR in the Gaussian linear channel [16]. Maximizing SINR is popular for waveform design when focusing on target detection problems. Based on the deterministic TIR model in [5], the optimal detection waveform is generalized to the multiple extended targets situation with detection constraints [9]. Unfortunately, it is far from practical to have exact information about TIR before detection, since TIR is highly sensitive to the line of sight [6]. To circumvent this difficulty, the TIR should be described by an uncertainty set or a random vector, then the robust waveform design should be taken into account (see [10], [17], [18], [19]). More concretely, maximizing worst-case SINR against the uncertainty TIR set is often considered to alleviate the effects caused by TIR disturbance, resulting in the minimax optimization problems [20]. When the stochastic TIR model is use, the waveform is optimized via maximizing the mathematical expectation of output

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SINR [17]. However, the mathematical expectation of output SINR only incorporates the first order statistical characteristic rather than the structure of distribution. Therefore, the waveform performance corresponding to this metric is not guaranteed for the complicated output SINR distribution.

B. Our Work

In this paper, we address the waveform design for extended targets with random TIR from a probabilistic view. The Peak-to-Average Ratio (PAR) constraint is imposed on waveform to meet the hardware demands of radar transmitter. A novel Probabilistically Robust Detection (PRD) waveform design metric is proposed. The corresponding algorithms are devised to optimize waveforms in high and low SINR situations, respectively. Numerical experiments are carried out to verify the effectiveness of the proposed waveform optimization algorithms and the PRD metric. Specifically, our work makes the following contributions:

1) *A novel PRD waveform design metric:* We have derived the closed forms of the mathematical expectation and variance of the output SINR with Gaussian random TIR, based on which, a novel PRD waveform design metric is proposed. The geometric interpretation and limit interpretation are given to show its rationality. Different from the conventional metric that maximizes the mathematical expectation of output SINR in [17], the newly proposed one takes the distribution structure of SINR into consideration. From this aspect, the PRD metric can be treated as a generalization of the metric in [17]. The PRD metric reduces to that in [17] for some special cases.

2) *Waveform covariance matrix design algorithm in high SINR situations:* We devise the algorithm to optimize waveform covariance matrix in high SINR situations. It is theoretically proved that the associated waveform covariance matrix design problem belongs to the quasi-convex fractional optimization problems. Therefore, the global optimal solution can be obtained based on Dinkelbach's algorithm [21] in polynomial time [22].

3) *Waveform covariance matrix design algorithm in low SINR situations:* Different from the high SINR cases, the optimal waveform covariance matrix in low SINR situations is hardly to be obtained, since the associated optimization problem is non-convex. Leveraging on Dinkelbach's algorithm and Minorization Maximization (MM) techniques [23], we propose the algorithm to find high qualified (sub-optimal) waveform covariance matrix solution through iterations. Further, the converge of the algorithm is proved theoretically.

The remainder of this paper is organized as follows. Section II establishes the signal model for extended targets and formulates the waveform design problem, where the PRD metric and its interpretation are included. Section III is devoted to the algorithms for waveform optimization based on the newly proposed metric. Further discussions for the proposed algorithms are presented in Section IV. Section V provides numerical experiments to test the performance of the proposed PRD metric as well as waveform optimization algorithms. Finally, conclusions are drawn in Section VI.

C. Notations

Throughout this paper, scalars are denoted by italic letters (e.g., a, A); vectors are denoted by bold italic lowercase letters (e.g., $\boldsymbol{a}$) and $\boldsymbol{a}(i)$ denotes the ith element of $\boldsymbol{a}$, $\mathbf{1}_N$ denotes the N-dimensional vector with all elements being 1. Matrices are denoted by bold italic capital letters (e.g., $\boldsymbol{A}$) and $\boldsymbol{A}(i,j)$ denotes the element in the ith row and jth column of $\boldsymbol{A}$. $\mathbf{I}_N$ is the unit matrix with size N. $\mathrm{tr}(\cdot)$ stands for the trace of a square matrix. Superscript $(\cdot)^{\mathrm{T}}$ and $(\cdot)^{\mathrm{H}}$ denote transpose and conjugate transpose, respectively. $\|\cdot\|_F$ is the Frobenius norm for matrices. $\mathcal{F}(\mathcal{P})$ and $\mathcal{V}(\mathcal{P})$ represent the feasible set and the optimal value of the optimization problem $\mathcal{P}$, respectively. The representation $\boldsymbol{A} \succ 0$ ($\boldsymbol{A} \succeq 0$) means $\boldsymbol{A}$ is positive definite (semi-definite). $\mathcal{CN}$ and χ^2 denote the complex Gaussian distribution and the chi-square distribution, respectively. $\mathbb{E}[x]$, $\mathbb{V}[x]$ and $\mathbb{D}[x]$ denote the expectation, variance and standard deviation of the random variable x, respectively.

II. Signal Model and Problem Formulation

A. Signal Model

Consider a high resolution radar transmitting discrete time waveform $\boldsymbol{s} = [\boldsymbol{s}(1), \boldsymbol{s}(2), \ldots, \boldsymbol{s}(L)] \in \mathbb{C}^L$ with L samples. Let $\boldsymbol{t} = [\boldsymbol{t}(1), \boldsymbol{t}(2), \ldots, \boldsymbol{t}(Q)] \in \mathbb{C}^Q$ represent the discrete samples of TIR, where Q is the number of range cells occupied by the extended target. Then, the digital received base band signal is given by (see the similar model in [5], [8], [9], [10], [11], [24], [25])

$$\boldsymbol{y} = \boldsymbol{T}\boldsymbol{s} + \boldsymbol{n}, \tag{1}$$

where $\boldsymbol{n} \sim \mathcal{CN}(\mathbf{0}, \boldsymbol{R}_n)$ stands for the signal independent interference at the receiver; $\boldsymbol{T} = \sum_{q=1}^{Q} \boldsymbol{t}(q)\boldsymbol{J}_{q-1}$ is the TIR matrix with $\boldsymbol{J}_q \in \mathbb{C}^{(L+Q-1)\times L}$ being the shift matrix [10], defined by

$$\boldsymbol{J}_q(m,n) = \begin{cases} 1, \text{ if } m-n=q \\ 0, \text{ if } m-n \neq q \end{cases}.$$

After a couple of algebraic manipulations [26], we can rewrite (1) as

$$\boldsymbol{y} = \boldsymbol{S}\boldsymbol{t} + \boldsymbol{n} \tag{2}$$

with $\boldsymbol{S} = [\boldsymbol{J}_0\boldsymbol{s}, \boldsymbol{J}_1\boldsymbol{s}, \ldots, \boldsymbol{J}_{Q-1}\boldsymbol{s}]$.

The output SINR γ at the receiver can be represented by

$$\gamma(\boldsymbol{s}) = \boldsymbol{t}^{\mathrm{H}}\boldsymbol{S}^{\mathrm{H}}\boldsymbol{R}_n^{-1}\boldsymbol{S}\boldsymbol{t}. \tag{3}$$

Since it is impossible to get the exact knowledge of TIR before detection, we assume TIR to be a random vector with $\boldsymbol{t} \sim \mathcal{CN}(\boldsymbol{t}_0, \boldsymbol{R}_t)$ (similar assumption can be also found in [5], [8], [11], [17]). Under this assumption, the parameter $\boldsymbol{t}_0$ describes the prior knowledge of TIR, while $\boldsymbol{R}_t$ describes the uncertainties as well as correlations of the elements in $\boldsymbol{t}$. Furthermore, we also assume that both $\boldsymbol{t}_0$ and $\boldsymbol{R}_t$ can be estimated from previous detection or target database through cognitive methods [27], [28].

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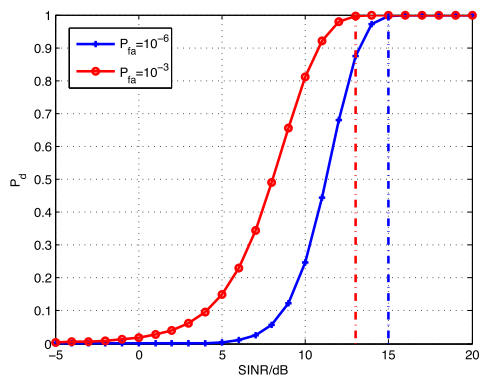

Fig. 1. Detection probability curve.

B. Waveform Design Metric

Our waveform design metric is inspired by the detection probability curve under the Gaussian channel, given by

$$P_d = \mathrm{MarcQ}\left(\sqrt{2\gamma(\boldsymbol{s})}, \sqrt{-2\log(P_{fa})}\right), \tag{4}$$

where $\mathrm{MarcQ}(\cdot,\cdot)$, P_d and P_{fa} are the Marcum-Q function, detection probability and false alarm rate [4], respectively. Fig. 1 plots the detection probability curves under different P_{fa}. It is indicated that the detection probability curve takes on the "S" shape, and the improvement of SINR brings few benefits to detection probability when SINR reaches a certain value. For example, the detection probability is about 99.7% for SINR = 13 dB with $P_{fa} = 10^{-3}$, thus there is no need to further increase the SINR value from the detection view.

In our consideration, the randomness of TIR leads to $\gamma(\boldsymbol{s})$ being a random variable. Thus, the actually achieved detection probability also becomes a random variable distributing along the detection probability curve. In most cases, radar practitioners want to detect the target of interest with a desired detection probability P_d^0 for a given P_{fa}. In the stochastic cases, one can maximize the probability of this event by designing transmit waveform. Noting the monotonic relations between $\gamma(\boldsymbol{s})$ and P_d in (4), we have the following waveform design principle

$$\max_{\boldsymbol{s}} \Pr\left\{\gamma(\boldsymbol{s}) \geq \gamma_0\right\}, \tag{5}$$

where γ_0 is the desired output SINR associated with P_d^0. Some statistical properties of $\gamma(\boldsymbol{s})$ are provided in the following proposition.

Proposition 1: Let $\boldsymbol{t} \sim \mathcal{CN}(\boldsymbol{t}_0, \boldsymbol{R}_t)$, then $\gamma(\boldsymbol{s}) = \frac{1}{2}\sum_{q=1}^{Q} \lambda_q \chi_2^2(2|\tilde{\boldsymbol{t}}_0(q)|^2)$ with

$$\mathbb{E}\left[\gamma(\boldsymbol{s})\right] = \mathrm{tr}\left(\boldsymbol{R}_t \boldsymbol{S}^{\mathrm{H}} \boldsymbol{R}_n^{-1} \boldsymbol{S}\right) + \boldsymbol{t}_0^{\mathrm{H}} \boldsymbol{S}^{\mathrm{H}} \boldsymbol{R}_n^{-1} \boldsymbol{S} \boldsymbol{t}_0$$

and

$$\mathbb{V}\left[\gamma(\boldsymbol{s})\right] = \mathrm{tr}\left(\boldsymbol{R}_t \boldsymbol{S}^{\mathrm{H}} \boldsymbol{R}_n^{-1} \boldsymbol{S} \boldsymbol{R}_t \boldsymbol{S}^{\mathrm{H}} \boldsymbol{R}_n^{-1} \boldsymbol{S}\right) + \boldsymbol{t}_0^{\mathrm{H}} \boldsymbol{S}^{\mathrm{H}} \boldsymbol{R}_n^{-1} \boldsymbol{S} \boldsymbol{R}_t \boldsymbol{S}^{\mathrm{H}} \boldsymbol{R}_n^{-1} \boldsymbol{S} \boldsymbol{t}_0,$$

where $\chi_2^2\left(2|\tilde{\boldsymbol{t}}_0(q)|^2\right)$ denotes the non-central χ^2 distribution with freedom 2 and non-central parameter $2|\tilde{\boldsymbol{t}}_0(q)|^2$; λ_q is the qth eigenvalue of matrix $\boldsymbol{R}_t^{1/2} \boldsymbol{S}^{\mathrm{H}} \boldsymbol{R}_n^{-1} \boldsymbol{S} \boldsymbol{R}_t^{1/2}$; $\tilde{\boldsymbol{t}}_0 = \boldsymbol{U} \boldsymbol{R}_t^{-1/2} \boldsymbol{t}_0$ with $\boldsymbol{U}$ formed by the eigenvector corresponding to λ_q.

Proof: See Appendix A. ■

Combing *Proposition 1* and results in [29], [30], we can write the closed form of (5) as

$$\Pr\left\{\gamma(\boldsymbol{s}) > \gamma_0\right\} = \frac{1}{2} + \frac{1}{\pi}\int_0^{+\infty} \frac{\sin\alpha}{u\beta} du, \tag{6}$$

where

$$\alpha = \sum_{q=1}^{Q}\left(\tan(\lambda_q u) + \frac{\left|\tilde{\boldsymbol{t}}_0(q)\right| \lambda_q u}{1+\lambda_q^2 u^2}\right) - \gamma_0 u,$$

$$\beta = \prod_{q=1}^{Q} \sqrt{1+\lambda_q^2 u^2}\, e^{\sum_{q=1}^{Q} \frac{|\tilde{\boldsymbol{t}}_0(q)|^2 \lambda_q^2 u^2}{1+\lambda_q^2 u^2}}.$$

Unfortunately, the expression of $\Pr\{\gamma(\boldsymbol{s}) \geq \gamma_0\}$ is so complicated that it is impossible to be optimized directly. To see this, we propose the following proxy function

$$\mathrm{f}(\boldsymbol{s}) = \frac{\mathbb{E}[\gamma(\boldsymbol{s})] - \gamma_0}{\mathbb{D}[\gamma(\boldsymbol{s})]} \tag{7}$$

to replace the objective in (5), where $\mathbb{D}[\gamma(\boldsymbol{s})] = \sqrt{\mathbb{V}[\gamma(\boldsymbol{s})]}$ is the standard deviation of $\gamma(\boldsymbol{s})$. To justify this replacement, we give interpretations from the perspective of geometry and limit as follows.

• Geometric interpretation. Geometrically, the objective in (5) can be regarded as the area formed by the probability density function of $\gamma(\boldsymbol{s})$ that exceeds γ_0 (see Fig. 2(a) for details). First, we consider high SINR situations ($\mathbb{E}[\gamma(\boldsymbol{s})] \geq \gamma_0$) depicted in Fig. 2(b). Under this situation, great gap between $\mathbb{E}[\gamma(\boldsymbol{s})]$ and γ_0, as well as small $\mathbb{D}[\gamma(\boldsymbol{s})]$ are preferred to get a high value of $\Pr\{\gamma(\boldsymbol{s}) \geq \gamma_0\}$. In Fig. 2(b), it is easy to check that $\boldsymbol{s}_1$ is a better choice compared with $\boldsymbol{s}_2$, and the inequality $\mathrm{f}(\boldsymbol{s}_1) > \mathrm{f}(\boldsymbol{s}_2)$ holds. On the other hand, consider low SINR situations ($\mathbb{E}[\gamma(\boldsymbol{s})] < \gamma_0$) depicted in Fig. 2(c). Obviously, narrow gap between $\mathbb{E}[\gamma(\boldsymbol{s})]$ and γ_0, as well as large $\mathbb{D}[\gamma(\boldsymbol{s})]$ will bring benefits. In this situation, the waveform $\boldsymbol{s}_2$ is better than $\boldsymbol{s}_1$, and it follows that $\mathrm{f}(\boldsymbol{s}_2) > \mathrm{f}(\boldsymbol{s}_1)$.

• Limit interpretation. According to the central-limit theorem [31], as Q approaches infinity, the distribution of $\gamma(\boldsymbol{s})$ is very close to the Gaussian distribution with mean $\mathbb{E}[\gamma(\boldsymbol{s})]$ and variance $\mathbb{V}[\gamma(\boldsymbol{s})]$. To this end, we have

$$\Pr\left\{\gamma(\boldsymbol{s}) \geq \gamma_0\right\} = \frac{1}{2}\left(1 - \mathrm{erf}\left(\frac{\gamma_0 - \mathbb{E}[\gamma(\boldsymbol{s})]}{2\mathbb{D}[\gamma(\mathrm{s})]}\right)\right), \tag{8}$$

where $\mathrm{erf}(\cdot)$ denotes the complementary error function [4]. Therefore, the waveform objective in (5) is equivalent to maximizing $\mathrm{f}(\boldsymbol{s})$ under the limit consideration.

Based on the analyses above, we take $\max_{\boldsymbol{s}} \mathrm{f}(\boldsymbol{s})$ as the PRD waveform design metric, given that the expression of $\mathrm{f}(\boldsymbol{s})$ is simpler compared with $\Pr\{\gamma(\boldsymbol{s}) \geq \gamma_0\}$. It should be noted that the design metric used in this paper can be regarded as a generalization of the random case in [17]. Particularly, the proposed design metric reduces to that in [17], if $\mathbb{V}[\gamma(\boldsymbol{s})]$ is constant and $\boldsymbol{t}_0$ is set to be $\boldsymbol{0}$.

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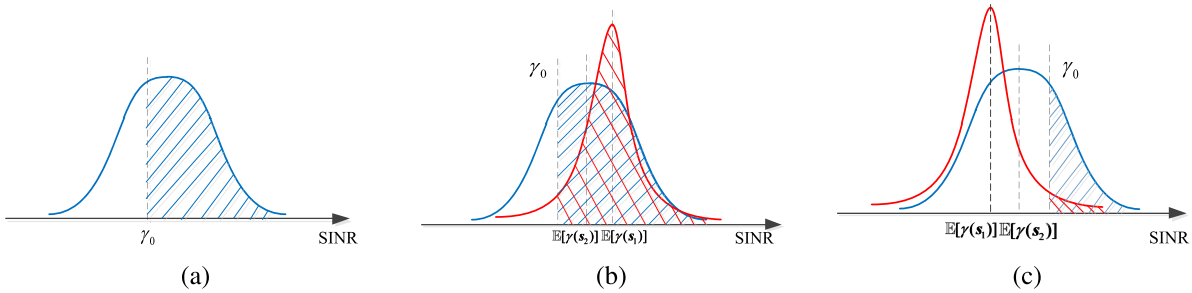

Fig. 2. Geometric interpretation for the proxy function: (a) original objective; (b) high SINR situations; (c) low SINR situations.

C. Waveform Constraint

To maximize the target detection abilities, the amplifier of radar always operates in the saturation region, and deep modulation of waveform amplitude is not allowed. For facilitating the hardware realization, the maximal amplitude modulation of the waveform should be controlled in practical situations. Consequently, the Peak-to-Average Ratio (PAR) constraint should be imposed on the waveform [32], [33], [34], given by

$$\|\boldsymbol{s}\|_2 = 1,\ |\boldsymbol{s}(i)|^2 \le \rho, i = 1, 2, \ldots, L, \tag{9}$$

where ρ is the parameter to control the maximal peak value in $\boldsymbol{s}$. The PAR constraint is a generalization of the Energy Constraint (EC) and Constant Modulus Constraint (CMC) constraint. Specifically, the PAR constraint reduces to the EC if ρ is set to be 1, meanwhile it denotes the CMC when $\rho = 1/L$.

Summarizing the analyses above, we formulate the PRD waveform design problem as

$$\mathcal{P}\begin{cases} \max\limits_{\boldsymbol{s}} \frac{\mathbb{E}[\gamma(\boldsymbol{s})]-\gamma_0}{\mathbb{D}[\gamma(\boldsymbol{s})]} \\ \text{s.t. } \|\boldsymbol{s}\|_2 = 1,\ |\boldsymbol{s}(i)|^2 \le \rho, i = 1, 2, \ldots, L. \end{cases} \tag{10}$$

At the end of this section, it is worth pointing out that the problem $\mathcal{P}$ is NP-hard [22] for the following reasons: i) the PAR constraint is non-convex; ii) $\mathrm{f}(\boldsymbol{s})$ is a non-linear fraction function with respect to $\boldsymbol{s}$; iii) $\mathbb{D}[\gamma(\boldsymbol{s})]$ is the square root of a quartic function with respect to $\boldsymbol{s}$. To tackle this problem, we are going to devise effective algorithms to find high qualified (sub-optimal) waveform in the next section.

III. Waveform Design Algorithm

Since $\mathcal{P}$ is difficult to solve directly, we investigate its Semi-Definite Relaxation (SDR) version [35] instead, given by

$$\tilde{\mathcal{P}}\begin{cases} \max\limits_{\boldsymbol{R}_s} \frac{\mathrm{tr}(\boldsymbol{R}_t\mathbf{M}(\boldsymbol{R}_s))+\boldsymbol{t}_0^{\mathrm{H}}\mathbf{M}(\boldsymbol{R}_s)\boldsymbol{t}_0-\gamma_0}{\sqrt{\mathrm{tr}(\boldsymbol{R}_t\mathbf{M}(\boldsymbol{R}_s)\boldsymbol{R}_t\mathbf{M}(\boldsymbol{R}_s))+\boldsymbol{t}_0^{\mathrm{H}}\mathbf{M}(\boldsymbol{R}_s)\boldsymbol{R}_t\mathbf{M}(\boldsymbol{R}_s)\boldsymbol{t}_0}} \\ \text{s.t. } \mathrm{tr}(\boldsymbol{R}_s) = 1, \boldsymbol{R}_s \succeq 0, \boldsymbol{R}_s(i,i) \le \rho, i = 1, 2, \ldots, L, \end{cases} \tag{11}$$

where $\boldsymbol{R}_s = \boldsymbol{s}\boldsymbol{s}^{\mathrm{H}}$ is the waveform covariance matrix and $\mathbf{M}(\boldsymbol{R}_s) = \boldsymbol{S}^H\boldsymbol{R}_n^{-1}\boldsymbol{S}$. It is easy to check that $\mathbf{M}(\boldsymbol{R}_s)$ is linear with respect to $\boldsymbol{R}_s$ with $\mathrm{tr}\left(\boldsymbol{J}_{m-1}^{\mathrm{H}}\boldsymbol{R}_n^{-1}\boldsymbol{J}_{n-1}\boldsymbol{R}_s\right)$ being the element in its mth row and nth column. Once the waveform $\boldsymbol{R}_s$ is optimized, the high-qualified $\boldsymbol{s}$ can be synthesized via randomization schemes [32]. To simplify the usage of symbols, we replace $\mathbf{M}(\boldsymbol{R}_s)$ by $\mathbf{M}$ in the sequel if no confusions are caused.

Note that $\tilde{\mathcal{P}}$ belongs to the fractional programming problems, thus we solve $\tilde{\mathcal{P}}$ through the Dinkelbach's algorithm [21]. In particular, we successively optimizing

$$\tilde{\mathcal{P}}^{(k)}\begin{cases} \max\limits_{\boldsymbol{R}_s} \quad \mathrm{tr}(\boldsymbol{R}_t\mathbf{M}) + \boldsymbol{t}_0^{\mathrm{H}}\mathbf{M}\boldsymbol{t}_0 - \gamma_0 \\ \qquad -f^{(k-1)}\sqrt{\mathrm{tr}(\boldsymbol{R}_t\mathbf{M}\boldsymbol{R}_t\mathbf{M}) + \boldsymbol{t}_0^{\mathrm{H}}\mathbf{M}\boldsymbol{R}_t\mathbf{M}\boldsymbol{t}_0} \\ \text{s.t. } \mathrm{tr}(\boldsymbol{R}_s)=1, \boldsymbol{R}_s(i,i)\le\rho, i=1,2,\ldots,L, \boldsymbol{R}_s\succeq 0. \end{cases} \tag{12}$$

starting from a feasible $\boldsymbol{R}_s^{(0)}$ until converges, where $f^{(k-1)}$ is the objective value in $\tilde{\mathcal{P}}$ associated with $\boldsymbol{R}_s^{(k-1)}$. Next, we devise algorithms to solve $\tilde{\mathcal{P}}^{(k)}$ in high and low SINR situations.

A. High SINR Situations

In the following discussions, the high SINR situation refers to the case that

$$\exists \boldsymbol{R}_{s_0} \in \mathcal{F}(\mathcal{P}),\ \text{ s.t. } \mathrm{tr}(\boldsymbol{R}_t\mathbf{M}_0) + \boldsymbol{t}_0^{\mathrm{H}}\mathbf{M}_0\boldsymbol{t}_0 - \gamma_0 \ge 0, \tag{13}$$

where $\mathbf{M}_0 = \mathbf{M}(\boldsymbol{R}_{s_0})$. Namely, there exists a feasible waveform (waveform covariance matrix) such that the maximal mathematical expectation of output SINR exceeds the desired γ_0.

Proposition 2: If the high SINR condition (13) meets, $\tilde{\mathcal{P}}$ is equivalent to the following quasi-convex optimization problem

$$\tilde{\mathcal{P}}_{\mathrm{h}}\begin{cases} \max\limits_{\boldsymbol{R}_s} \frac{\mathrm{tr}(\boldsymbol{R}_t\mathbf{M})+\boldsymbol{t}_0^{\mathrm{H}}\mathbf{M}\boldsymbol{t}_0-\gamma_0}{\sqrt{\mathrm{tr}(\boldsymbol{R}_t\mathbf{M}\boldsymbol{R}_t\boldsymbol{M})+\boldsymbol{t}_0^{\mathrm{H}}\mathbf{M}\boldsymbol{R}_t\mathbf{M}\boldsymbol{t}_0}} \\ \text{s.t. } \mathrm{tr}(\boldsymbol{R}_s) = 1, \boldsymbol{R}_s(i,i) \le \rho, i = 1, 2, \ldots, L, \\ \quad \boldsymbol{R}_s \succeq 0, \mathrm{tr}(\boldsymbol{R}_t\boldsymbol{M}) + \boldsymbol{t}_0^{\mathrm{H}}\boldsymbol{M}\boldsymbol{t}_0 - \gamma_0 \ge 0. \end{cases}$$

Proof: See Appendix B. ■

As an immediate consequence of *Proposition 2*, the optimal solution of $\tilde{\mathcal{P}}^{(k)}$ in high SINR situations can be equivalently obtained by solving

$$\tilde{\mathcal{P}}_{\mathrm{h}}^{(k)}\begin{cases} \max\limits_{\boldsymbol{R}_s} \quad \mathrm{tr}(\boldsymbol{R}_t\mathbf{M}) + \boldsymbol{t}_0^{\mathrm{H}}\mathbf{M}\boldsymbol{t}_0 - \gamma_0 \\ \qquad -f^{(k-1)}\sqrt{\mathrm{tr}(\boldsymbol{R}_t\mathbf{M}\boldsymbol{R}_t\mathbf{M}) + \boldsymbol{t}_0^{\mathrm{H}}\mathbf{M}\boldsymbol{R}_t\mathbf{M}\boldsymbol{t}_0} \\ \text{s.t. } \mathrm{tr}(\boldsymbol{R}_s) = 1, \boldsymbol{R}_s(i,i) \le \rho, i = 1, 2, \ldots, L, \\ \quad \boldsymbol{R}_s \succeq 0, \mathrm{tr}(\boldsymbol{R}_t\mathbf{M}) + \boldsymbol{t}_0^{\mathrm{H}}\mathbf{M}\boldsymbol{t}_0 - \gamma_0 \ge 0. \end{cases} \tag{14}$$

Note that $\sqrt{\mathrm{tr}(\boldsymbol{R}_t\boldsymbol{M}\boldsymbol{R}_t\boldsymbol{M}) + \boldsymbol{t}_0^{\mathrm{H}}\boldsymbol{M}\boldsymbol{R}_t\boldsymbol{M}\boldsymbol{t}_0}$ is convex with respect to $\boldsymbol{R}_s$ (see the proof in Appendix B), and the constraint $\mathrm{tr}(\boldsymbol{R}_t\boldsymbol{M}) + \boldsymbol{t}_0^{\mathrm{H}}\boldsymbol{M}\boldsymbol{t}_0 - \gamma_0 \ge 0$ guarantees the inequality $f^{(k-1)} \ge 0$ for any k. Hence, $\tilde{\mathcal{P}}_{\mathrm{h}}^{(k)}$ describes a convex optimization problem that maximizes a concave objective over a convex set. More precisely, $\tilde{\mathcal{P}}_{\mathrm{h}}^{(k)}$ is a Semi-Definite Programming (SDP) problem that can be solved with at most $O(L^9)$ floating

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operations [22], [36], [37], where one floating-point operation stands for one addition or multiplication of two floating-point numbers.

Algorithm 1 summarizes the procedure to tackle $\tilde{\mathcal{P}}$ in high SINR situations. Here, we briefly analyze the convergence property of **Algorithm 1**. Note that $\tilde{\mathcal{P}}_{\rm h}$ is a quasi-convex optimization problem. As proved in [21], the Dinkelbach's algorithm reaches the global optimal point for the quasi-convex optimization problem. Therefore, convergences of **Algorithm 1** are guaranteed by the quasi-convexity of $\tilde{\mathcal{P}}_{\rm h}$.

B. Low SINR Situations

This subsection is devoted to the algorithm for $\tilde{\mathcal{P}}^{(k)}$ in low SINR situations where

$$\forall \boldsymbol{R}_s \in \mathcal{F}(\mathcal{P}),\ \text{s.t.}\ \operatorname{tr}(\boldsymbol{R}_t\boldsymbol{M}) + \boldsymbol{t}_0^{\rm H}\mathbf{M}\boldsymbol{t}_0 - \gamma_0 < 0. \quad (15)$$

Namely, for any feasible waveform (or waveform covariance matrix), the mathematical expectation of output SINR is always lower than the desired γ_0. It is worth noting that the problem $\tilde{\mathcal{P}}^{(k)}$ in low SINR situations is essentially different from that in the high SINR situations, although they share almost the same structure. For the high SINR situation, the condition $f^{(k-1)} \geq 0$ guarantees the convexity of $\tilde{\mathcal{P}}^{(k)}$. By contrast, $\tilde{\mathcal{P}}^{(k)}$ is non-convex due to the low SINR situation ($f^{(k-1)} < 0$). To see this, we tackle $\tilde{\mathcal{P}}^{(k)}$ in low SINR situations based on MM techniques (more details about the MM techniques can be found in [23], [33], [38] and references therein).

Note that convex function is minorized by its supporting hyperplane [22], then we have

$$\begin{aligned}&\sqrt{\operatorname{tr}(\boldsymbol{R}_t\mathbf{M}\boldsymbol{R}_t\mathbf{M}) + \boldsymbol{t}_0^{\rm H}\mathbf{M}\boldsymbol{R}_t\mathbf{M}\boldsymbol{t}_0}\\ &\quad\geq \sqrt{\operatorname{tr}\left(\boldsymbol{R}_t\mathbf{M}^{(k)}\boldsymbol{R}_t\mathbf{M}^{(k)}\right) + \boldsymbol{t}_0^{\rm H}\mathbf{M}^{(k)}\boldsymbol{R}_t\mathbf{M}^{(k)}\boldsymbol{t}_0}\\ &\qquad + \operatorname{tr}\left(\boldsymbol{G}^{(k)}(\mathbf{M}-\mathbf{M}^{(k)})\right),\end{aligned} \quad (16)$$

where $\mathbf{M}^{(k)} = \mathbf{M}\left(\boldsymbol{R}_s^{(k)}\right)$ and $\boldsymbol{G}^{(k)}$ is the gradient of $\sqrt{\operatorname{tr}(\boldsymbol{R}_t\mathbf{M}\boldsymbol{R}_t\mathbf{M}) + \boldsymbol{t}_0^{\rm H}\mathbf{M}\boldsymbol{R}_t\mathbf{M}\boldsymbol{t}_0}$ at $\mathbf{M}^{(k)}$ with

$$\boldsymbol{G}^{(k)} = \frac{\boldsymbol{R}_t\mathbf{M}^{(k)}\left(\boldsymbol{R}_t + \boldsymbol{t}_0\boldsymbol{t}_0^{\rm H}\right)}{\sqrt{\operatorname{tr}\left(\boldsymbol{R}_t\mathbf{M}^{(k)}\boldsymbol{R}_t\mathbf{M}^{(k)}\right) + \boldsymbol{t}_0^{\rm H}\mathbf{M}^{(k)}\boldsymbol{R}_t\mathbf{M}^{(k)}\boldsymbol{t}_0}}. \quad (17)$$

Here, the principle of complex-valued matrix derivatives in [39] and Hermitian properties of $\mathbf{M}$ are used to derive (17).

At the vth iteration of the MM techniques, define $\mathbf{M}^{(k,v)} = \mathbf{M}\left(\boldsymbol{R}_s^{(k,v)}\right)$. It follows from (16) that

$$\mathrm{g}^{(k,v)}(\mathbf{M}) = \operatorname{tr}\left(\left(\boldsymbol{R}_t + \boldsymbol{t}_0\boldsymbol{t}_0^{\rm H} - f^{(k-1)}\boldsymbol{G}^{(k,v)}\right)\mathbf{M}\right) + c^{(k,v)}$$

provides a minorizer of the objective in $\tilde{\mathcal{P}}^{(k)}$ at $\mathbf{M}^{(k,v)}$, where $\boldsymbol{G}^{(k,v)}$ is calculated with (17) via replacing $\mathbf{M}^{(k)}$ by $\mathbf{M}^{(k,v)}$, and $c^{(k,v)} = -\gamma_0 + f^{(k-1)}\operatorname{tr}(\boldsymbol{G}^{(k,v)}\mathbf{M}^{(k,v)}) - f^{(k-1)}\sqrt{\operatorname{tr}\left(\boldsymbol{R}_t\mathbf{M}^{(k,v)}\boldsymbol{R}_t\mathbf{M}^{(k,v)}\right) + \boldsymbol{t}_0^{\rm H}\mathbf{M}^{(k,v)}\boldsymbol{R}_t\mathbf{M}^{(k,v)}\boldsymbol{t}_0}$ is a constant unrelated to $\boldsymbol{R}_s$.

Algorithm 1: Waveform Covariance Matrix Design Under High SINR Situations.

Input: $\boldsymbol{t}_0,\ \boldsymbol{R}_t,\ \boldsymbol{R}_n,\ \gamma_0,\ \boldsymbol{R}_s^{(0)},\ f^{(0)}$.
Output: $\boldsymbol{R}_s$.
Start: Set $k = 0$.
repeat
  Solve the convex problem $\tilde{\mathcal{P}}_{\rm h}^{(k+1)}$ by CVX and record the corresponding solution as $\boldsymbol{R}_s^{(k+1)}$;
  Set $k = k + 1$;
until *Converges*;
Set $\boldsymbol{R}_s = \boldsymbol{R}_s^{(k)}$.

According to MM techniques, the sub-optimal solution of $\tilde{\mathcal{P}}^{(k)}$ in the low SINR situations can be achieved by iteratively solving

$$\tilde{\mathcal{P}}_1^{(k,v)}\begin{cases}\max\limits_{\boldsymbol{R}_s}\ \ \mathrm{g}^{(k,v)}(\boldsymbol{M})\\ \text{s.t.}\ \operatorname{tr}(\boldsymbol{R}_s) = 1,\ \boldsymbol{R}_s \succeq 0,\\ \qquad \boldsymbol{R}_s(i,i) \leq \rho, i = 1,2,\ldots,L,\end{cases} \quad (18)$$

until converges. Obviously, $\tilde{\mathcal{P}}_1^{(k,v)}$ is also a SDP problem [22] that can be efficiently solved through interior-point algorithm with at most $O(L^{4.5})$ floating operations [35], [36].

Algorithm 2 lists the whole procedure to solve $\tilde{\mathcal{P}}$ in low SINR situations for better understanding. The following analyses provide an insight to the convergence of **Algorithm 2**. According to the monotonically increasing properties of MM techniques, we have

$$\mathcal{V}(\tilde{\mathcal{P}}_1^{(k,v)}) \leq \mathcal{V}(\tilde{\mathcal{P}}_1^{(k,v+1)}). \quad (19)$$

On the other hand, we observe from **Algorithm 2** that $\boldsymbol{R}_s^{(k,0)} = \boldsymbol{R}_s^{(k)}$, which means

$$\mathcal{V}(\tilde{\mathcal{P}}_1^{(k,0)}) = 0. \quad (20)$$

Denote by $\boldsymbol{R}_s^{(k)}$ the solution of $\tilde{\mathcal{P}}^{(k)}$ output by MM, then we have

$$\begin{aligned}&\operatorname{tr}(\boldsymbol{R}_{\rm t}\mathbf{M}^{(\rm k)}) + \boldsymbol{t}_0^{\rm H}\mathbf{M}^{(\rm k)}\boldsymbol{t}_0 - \gamma_0\\ &\quad - f^{(k-1)}\sqrt{\operatorname{tr}\left(\boldsymbol{R}_t\mathbf{M}^{(k)}\boldsymbol{R}_t\mathbf{M}^{(k)}\right) + \boldsymbol{t}_0^{\rm H}\mathbf{M}^{(k)}\boldsymbol{R}_t\mathbf{M}^{(k)}\boldsymbol{t}_0} \geq 0.\end{aligned} \quad (21)$$

As an immediate consequence of (21), we get

$$\frac{\operatorname{tr}\left(\boldsymbol{R}_t\mathbf{M}^{(k)}\right) + \boldsymbol{t}_0^{\rm H}\mathbf{M}^{(k)}\boldsymbol{t}_0 - \gamma_0}{\sqrt{\operatorname{tr}(\boldsymbol{R}_t\mathbf{M}^{(k)}\boldsymbol{R}_t\mathbf{M}^{(k)}) + \boldsymbol{t}_0^{\rm H}\mathbf{M}^{(k)}\boldsymbol{R}_t\mathbf{M}^{(k)}\boldsymbol{t}_0}} \geq f^{(k-1)}, \quad (22)$$

which demonstrates that **Algorithm 2** forms a non-decreasing objective value sequence during iterations. Besides, $\mathcal{V}(\tilde{\mathcal{P}})$ is upper bounded due to the constraint $\operatorname{tr}(\boldsymbol{R}_s) = 1$. Combing the corresponding non-decreasing and upper bounded properties, the convergence of **Algorithm 2** is proved.

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Algorithm 2: Waveform Covariance Matrix Design Under Low SINR Situations.

Input: $\boldsymbol{t}_0,\ \boldsymbol{R}_t,\ \boldsymbol{R}_n,\ \gamma_0,\ \boldsymbol{R}_s^{(0)},\ f^{(0)}$.
Output: $\boldsymbol{R}_s$.
Start: Set $k=0$.
repeat
  Set $v=0$, $\boldsymbol{R}_s^{(k,0)}=\boldsymbol{R}_s^{(k)}$.
  repeat
    Solve the SDP problem $\tilde{\mathcal{P}}_1^{(k,v)}$ and record the corresponding solution as $\boldsymbol{R}_s^{(k,v+1)}$;
    Set $v=v+1$;
  until *Converges*;
  Set $\boldsymbol{R}_s^{(k+1)}=\boldsymbol{R}_s^{(k,v+1)}$, $k=k+1$;
until *Converges*;
Set $\boldsymbol{R}_s=\boldsymbol{R}_s^{(k)}$.

C. Synthesizing Waveform From $\boldsymbol{R}_s$

Once the optimal (sub-optimal) waveform covariance matrix $\boldsymbol{R}_s$ is obtained, we need to synthesize the feasible transmit waveform from $\boldsymbol{R}_s$.

The synthesis schemes are similar to that proposed in [32]. Let us introduce it briefly. If $\boldsymbol{R}_s$ is rank-one, the optimal (sub-optimal) waveform $\boldsymbol{s}^*$ can be directly synthesized with $\boldsymbol{R}_s=\boldsymbol{s}^*(\boldsymbol{s}^*)^{\mathrm{H}}$. Otherwise, we generate N waveforms complying with the PAR constraint according to randomization schemes in [32], then choose the one which maximizes the function value in (7).

IV. Further Discussions on Waveform Design Algorithm

A. Initialization and Algorithm Summarizing

It is known that the Dinkelbach's algorithm requires a feasible initialization. In this paper, we initialize $\boldsymbol{R}_s^{(0)}$ with the optimal solution of the following problem

$$\tilde{\mathcal{P}}_0\begin{cases}\max\limits_{\boldsymbol{R}_s}\ \mathrm{tr}(\boldsymbol{R}_t\mathbf{M})+\boldsymbol{t}_0^{\mathrm{H}}\mathbf{M}\boldsymbol{t}_0-\gamma_0\\ \text{s.t. } \mathrm{tr}(\boldsymbol{R}_s)=1,\boldsymbol{R}_s\succeq 0,\boldsymbol{R}_s(i,i)\le\rho, i=1,2,\ldots,L.\end{cases}\tag{23}$$

It should be noted that $\mathcal{V}(\tilde{\mathcal{P}}_0)$ distinguishes the SINR situation where radar operates. In detail, the high SINR condition (13) meets, if $\mathcal{V}(\tilde{\mathcal{P}}_0)\ge 0$. Otherwise, the radar operates in low SINR situations.

Combing the initialization schemes and the algorithms provided in Section III, we summarize the algorithm for PRD waveform design in **Algorithm 3**.

B. Detection Rules

This subsection discusses the possible detection rules with the optimized waveform. Consider the amplitude detector with

$$\left|\boldsymbol{w}^{\mathrm{H}}\boldsymbol{y}\right|\ge T,\tag{24}$$

where $\boldsymbol{w}$ is the filter at the receiver; $\boldsymbol{y}$ is the received returns associated with the optimized waveform; T is the threshold determined by the false alarm rate.

Algorithm 3: Algorithm for PRD Waveform Design.

Input: $\boldsymbol{t}_0,\ \boldsymbol{R}_t,\ \boldsymbol{R}_n,\ \gamma_0,\ \boldsymbol{R}_s^{(0)},\ f^{(0)}$.
Output: $\boldsymbol{s}^*$.
Initialization: Set $k=0$; initialize $\boldsymbol{R}_s^{(0)}$ and $f^{(0)}$ with the optimal solution of $\tilde{\mathcal{P}}_0$ and $\mathcal{V}(\tilde{\mathcal{P}}_0)$, respectively.
if $f^{(0)}\ge 0$ **then**
  Call **Algorithm 1** to optimize $\boldsymbol{R}_s$;
else
  Call **Algorithm 2** to optimize $\boldsymbol{R}_s$;
end
Synthesize waveform $\boldsymbol{s}^*$ from $\boldsymbol{R}_s$.

As shown in [4], the whitened match filter $\boldsymbol{w}=\boldsymbol{R}_n^{-1}\boldsymbol{S}\boldsymbol{t}$ is the best choice if $\boldsymbol{t}$ is deterministic and known. However, $\boldsymbol{t}$ is random and unknown in our consideration. A possible way to deal with this problem is utilizing $\boldsymbol{w}=\boldsymbol{R}_n^{-1}\boldsymbol{S}\hat{\boldsymbol{t}}$ instead [40], where $\hat{\boldsymbol{t}}$ is an estimation of $\boldsymbol{t}$ from the observing data $\boldsymbol{y}$. To further simplify the estimation procedure, one can also set $\boldsymbol{w}=\boldsymbol{R}_n^{-1}\boldsymbol{S}\boldsymbol{t}_0$ without using $\boldsymbol{y}$, since $\boldsymbol{t}_0$ is the maximum likelihood estimation of the prior knowledge $\mathcal{CN}(\boldsymbol{t}_0,\boldsymbol{R}_t)$.

V. Numerical Experiments

In this section, numerical experiments are provided to evaluate the PRD waveform design metric and corresponding waveform design algorithms.

Unless otherwise specified, in the following experiments, we consider a radar system transmitting the 0.6 us pulse waveform with 100 MHz sampling rate (1.5 m per range cell), resulting in $L=60$ samples in a pulse. The signal independent interference is assumed to be white with $\boldsymbol{R}_n=\sigma^2\mathbf{I}_{L+Q-1}$, where σ^2 is the interference level. The false alarm rate for the system is assumed to be 10^{-6}. According to the results in Fig. 1, the desired SINR γ_0 is set to be 15 dB to achieve relatively high detection probability. We terminate the iterations when the relative changes for the objective value are less than 10^{-4}. $N=1000$ samples are generated to synthesize waveform from $\boldsymbol{R}_s$ via randomization schemes.

Two extended targets are illustrated as examples in the following experiments. The first one (Target 1) is the typical fighter with 15 meters long, occupying $Q_1=10$ range cells. Its TIR information is given by

$$\begin{cases}\boldsymbol{t}_{0,1}(q)=\left(\cos\left(0.6\pi q-1.8\pi\right)+1\right)e^{-|q-2|+\mathrm{j}\pi(q-4)/6},\\ \boldsymbol{R}_{t,1}(m,n)=1.2e^{-(m-n)^2}+0.2e^{\mathrm{j}2(m-n)}.\end{cases}\tag{25}$$

The other one (Target 2) is the airliner with 38 meters long, occupying $Q_2=25$ rang cells. The TIR information of the airliner is given by

$$\begin{cases}\boldsymbol{t}_{0,2}=0.1\times\mathbf{1}_{Q_2},\\ \boldsymbol{R}_{t,2}(m,n)=0.2e^{\mathrm{j}2(m-n)}+0.1e^{\mathrm{j}4(m-n)}+0.3e^{\mathrm{j}6(m-n)}.\end{cases}\tag{26}$$

A. Convergence Properties

In this subsection, we conduct experiments to verify the convergence of the proposed algorithm.

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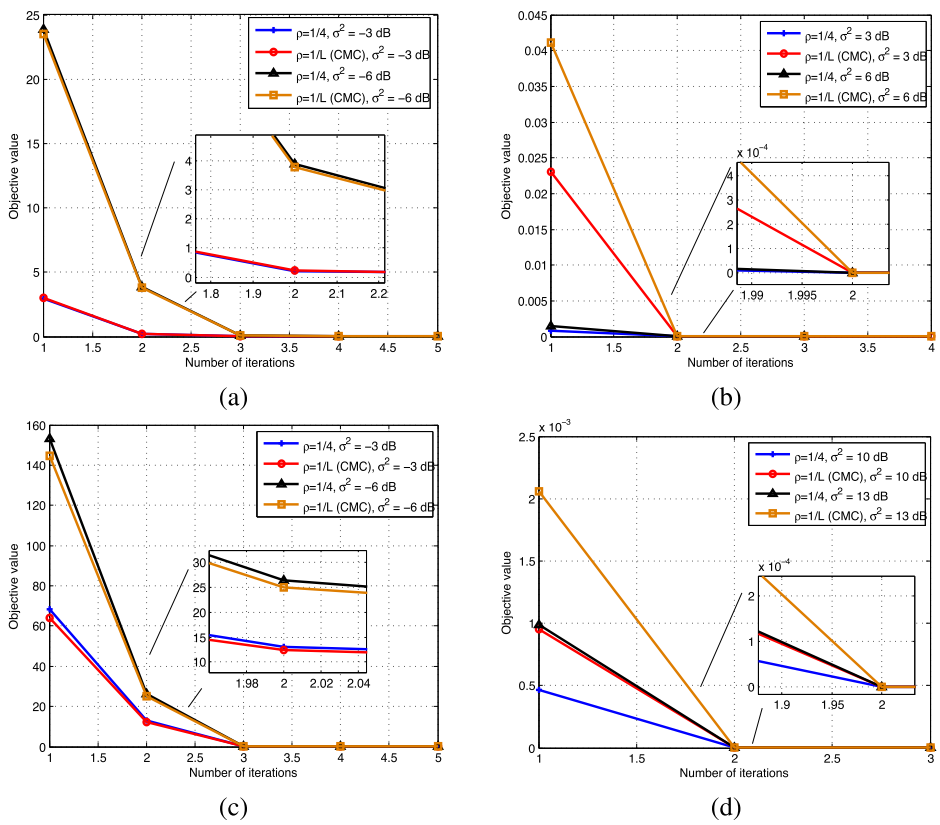

Fig. 3. Iteration curves: (a) Target 1, high SINR situations; (b) Target 1, low SINR situations; (c) Target 2, high SINR situations; (d) Target 2, low SINR situations.

The objective value for the Dinkelbach's iterations are shown in Fig. 3. It is indicated that the objective value decreases to 0 within several iterations, which is consistent with our theoretical analyses. These results demonstrate the convergence of **Algorithm 3** numerically. A careful inspection of the objective value in Fig. 3(b) and Fig. 3(d) reveals that, for the low SINR situations, the initialization is very close to the sub-optimal solution since the associated objective for initialization is almost equal to 0.

B. Synthesizing Waveform

The following experiment investigate the synthesizing procedure from the optimized waveform covariance matrix.

The power curves of synthesized waveform under different parameters are plotted in Fig. 4. It can be observed from the figure that the designed waveform meets the PAR constraint. Particularly, for Target 1 with $\rho = 1/20$, the maximal power of the designed waveform is 0.031 (lower than 1/20); for Target 2 with $\rho = 1/L$ (CMC), the power of the designed waveform is constant.

The objective value losses caused by the waveform synthesizing are illustrated in Fig. 5, where the dotted line and solid line stands for the objective value achieved by the optimal (sub-optimal) waveform covariance matrix and waveform, respectively. One can find that the parameter ρ takes few effects on the objective value for the reason that objective value curve is almost a straight line for different ρ. Moreover, the maximal loss

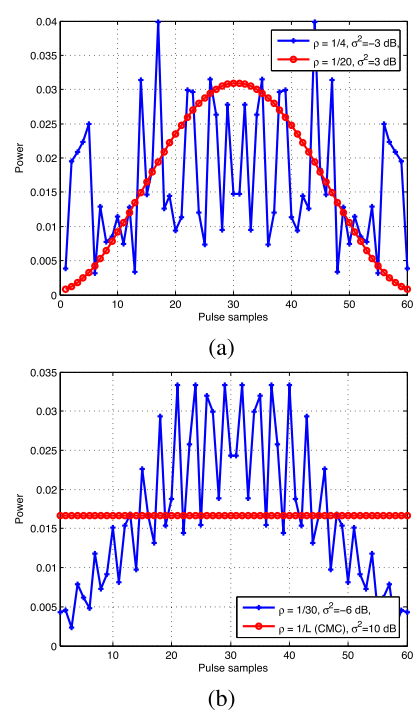

Fig. 4. Power of designed waveforms: (a) Target 1; (b) Target 2.

caused by waveform synthesizing is less than 0.1, which verifies the effectiveness of the waveform synthesizing procedure. Combing the results in Fig. 4 and Fig. 5, we get that the parameter

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TABLE I
STATISTICAL CHARACTERISTICS COMPARISONS FOR DIFFERENT DESIGN METRICS

	σ^2	Metric in [17]			Our metric		
		$\mathbb{E}[\gamma]$ /dB	$\mathbb{D}[\gamma]$ /dB	$\frac{\mathbb{E}[\gamma]-\gamma_0}{\mathbb{D}[\gamma]}$	$\mathbb{E}[\gamma]$ /dB	$\mathbb{D}[\gamma]$ /dB	$\frac{\mathbb{E}[\gamma]-\gamma_0}{\mathbb{D}[\gamma]}$
Target 1	-10 dB	24.41	24.32	0.91	20.90	15.90	2.35
	-5 dB	19.41	19.32	0.65	17.76	13.67	1.20
	0 dB	14.42	14.32	-0.15	14.42	14.32	-0.15
	5 dB	9.42	9.32	-2.68	9.41	9.33	-2.67
Target 2	-10 dB	32.26	32.32	0.99	21.01	16.96	1.90
	0 dB	22.26	22.22	0.82	19.83	17.54	1.14
	10 dB	12.26	12.22	-0.89	12.26	12.22	-0.89
	20 dB	2.26	2.22	-17.96	2.26	2.22	-17.96

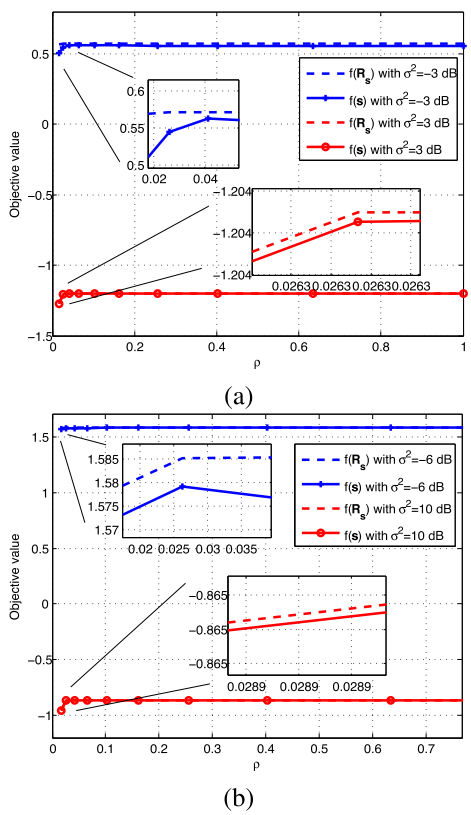

Fig. 5. Synthesizing losses: (a) Target 1; (b) Target 2.

ρ controls the modulation depth of waveform amplitude, but has little impact on waveform performance from the view of objective value.

C. Waveform Performance and Comparisons

In this subsection, experiments are provided to evaluate the waveform performance designed by our PRD metric, where the performance corresponding to the metric in [17] is also illustrated as benchmarks for comparisons. In the following experiments, we set $\rho = 1/L$ (CMC) for the sake that the constant modulus waveforms (such as phase coded waveform, frequency modulated waveform) are popular in practical situations.

Fig. 6 and Fig. 7 illustrate the output SINR distribution histogram for Target 1 and Target 2, respectively. In detail, we take the optimized waveform as transmit waveform and randomly generate 1000 TIRs (Monte Carlo trials) from $\mathcal{CN}(\boldsymbol{t}_0, \boldsymbol{R}_t)$, then calculate the distribution of the achieved output SINR. The statistical characteristics of the associated SINR distribution are listed in Table I. It is observed from the figure that, for the metric in [17], the shape of output SINR distribution does not change except for moving along the SINR axis when σ^2 varies. On the contrary, the parameter σ^2 affects the structure of output SINR distribution for the PRD metric. Precisely, the two metrics behave very similarly in low SINR situations (see the results in Fig. 6(c), Fig. 6(d), Fig. 7(c) and Fig. 7(d) and the corresponding data in Table I), but the output SINR associated with the PRD metric exhibits a much smaller variance in high SINR situations (see the results in Fig. 6(a), Fig. 6(b), Fig. 7(a) and Fig. 7(b) and the corresponding data in Table I). An inspection of the data in Table I reveals that, for the metric in [17], $\mathbb{E}[\gamma]$ and $\mathbb{D}[\gamma]$ increase with the same proportion when σ^2 decreases. While for the PRD metric, $\mathbb{E}[\gamma]$ also increases when σ^2 decreases, but the increment of $\mathbb{D}[\gamma]$ is controlled. Therefore, the PRD metric achieves a much higher value of $\frac{\mathbb{E}[\gamma]-\gamma_0}{\mathbb{D}[\gamma]}$ in high SINR situations.

Fig. 8 and Fig. 9 depict the waveform Power Spectral Density (PSD) [41] curves associated with the two metrics, to have further insight to the properties of designed waveforms. It is indicated that the waveform PSD for the metric in [17] always places its main peak at the maximal point of target PSD, regardless of the value of σ^2. This phenomenon explains the reason why σ^2 does not affect the shape of output SINR distribution for the metric in [17]. By contrast, in high SINR situations, the waveform PSD for the PRD metric tends to capture the structure of target PSD instead of forming a single peak at the maximal point of target PSD (see the results in Fig. 8(a), Fig. 8(b), Fig. 9(a) and Fig. 9(b)). This property enhance the robustness of the designed waveform, even though it suffers from some losses of $\mathbb{E}[\gamma]$ compared with the metric in [17]. In low SINR situations (Fig. 8(c), Fig. 8(d), Fig. 9(c) and Fig. 9(d)), the waveform PSD associated with the PRD metric is almost the same as that associated with the metric [17], which is consistent with the corresponding results in Fig. 6, Fig. 7 and Table I.

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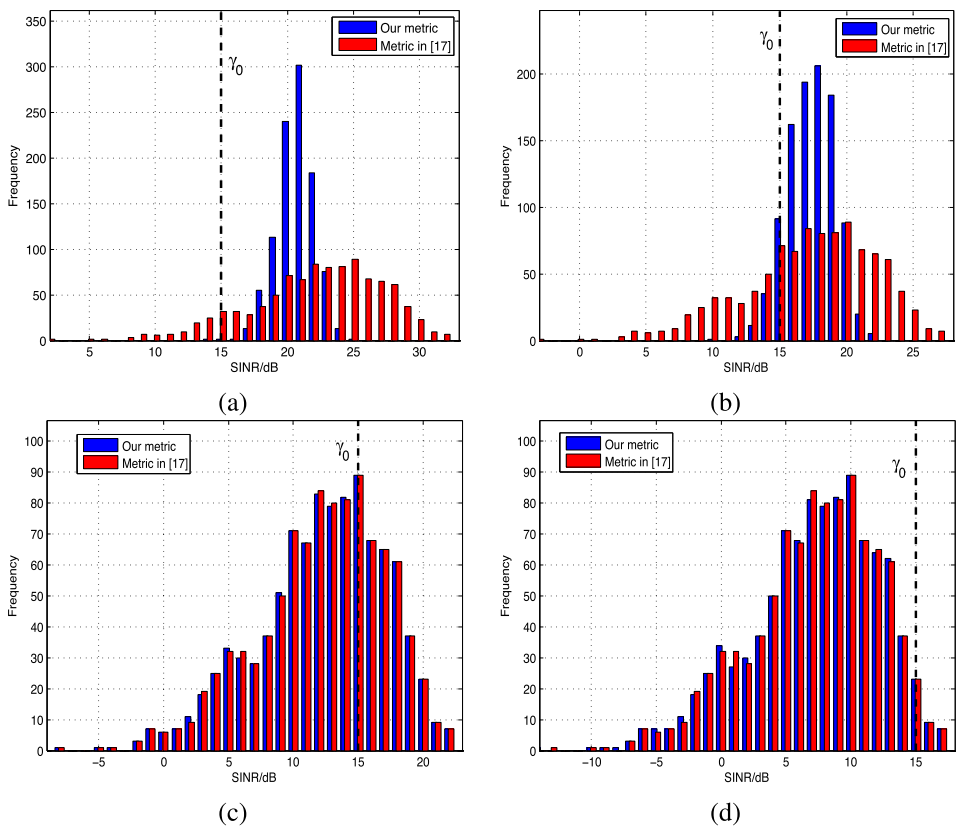

Fig. 6. SINR histogram for Target 1: (a) $\sigma^2 = -10$ dB; (b) $\sigma^2 = -5$ dB; (c) $\sigma^2 = 0$ dB; (d) $\sigma^2 = 5$ dB.

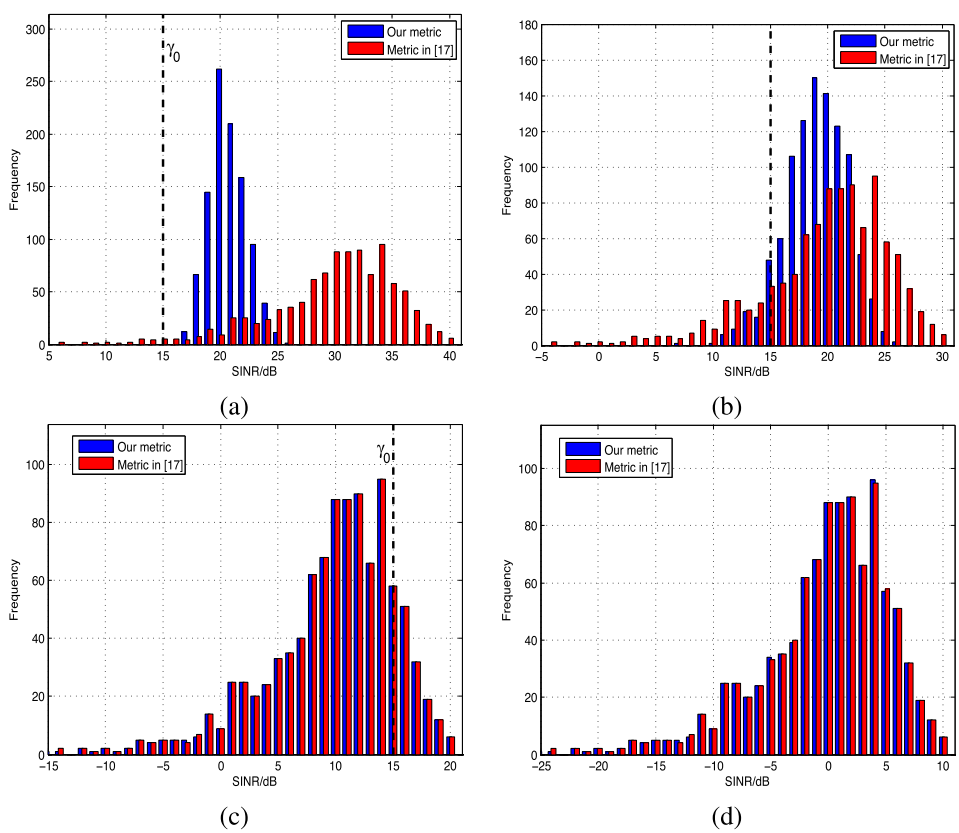

Fig. 7. SINR histogram for Target 2: (a) $\sigma^2 = -10$ dB; (b) $\sigma^2 = 0$ dB; (c) $\sigma^2 = 10$ dB; (d) $\sigma^2 = 20$ dB.

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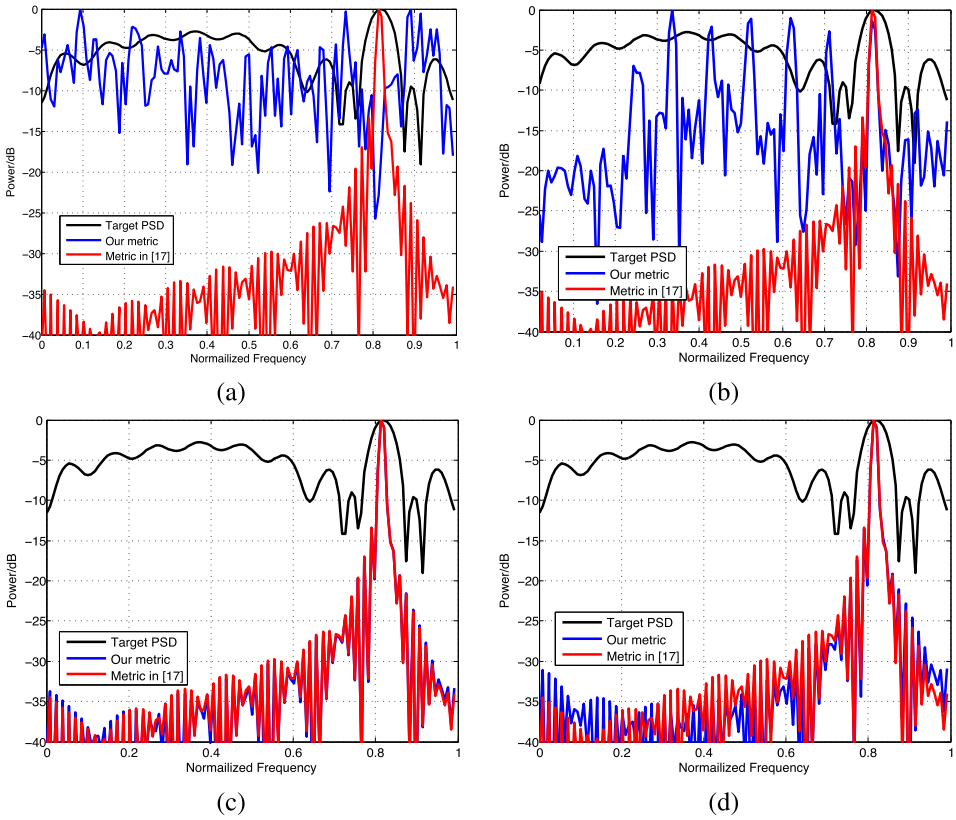

Fig. 8. Waveform PSD for Target 1: (a) $\sigma^2 = -10$ dB; (b) $\sigma^2 = -5$ dB; (c) $\sigma^2 = 0$ dB; (d) $\sigma^2 = 5$ dB.

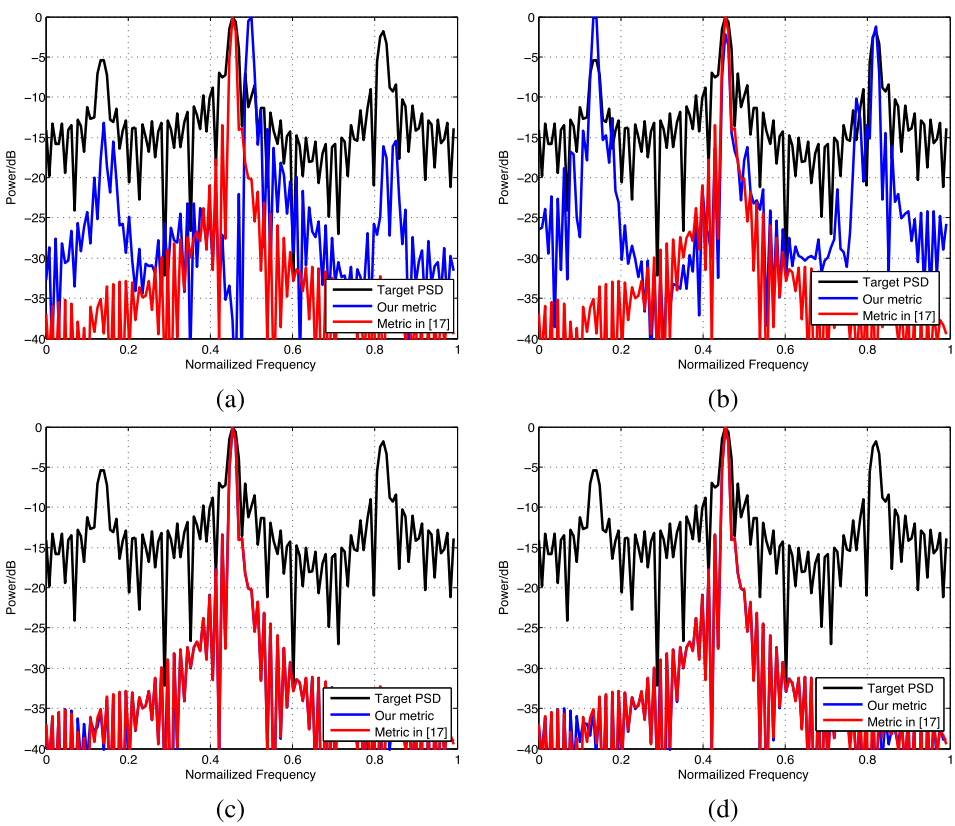

Fig. 9. Waveform PSD for Target 1: (a) $\sigma^2 = -10$ dB; (b) $\sigma^2 = -5$ dB; (c) $\sigma^2 = 0$ dB; (d) $\sigma^2 = 5$ dB.

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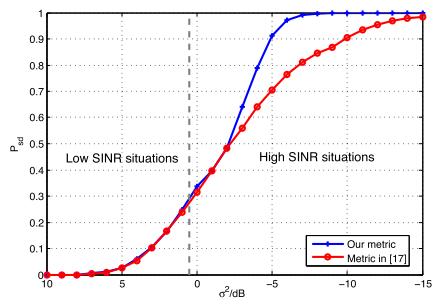
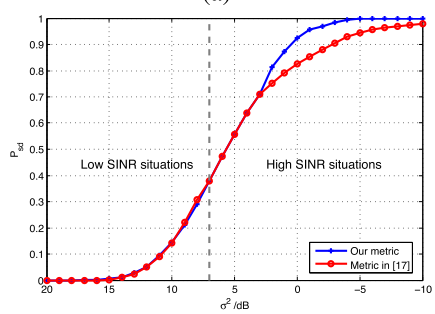


(a)



(b)

Fig. 10. P_{sd} versus σ^2: (a) Target 1; (b) Target 2.

The quantitatively evaluate the two design metrics from the view of detection performance, we define stable detection probability as

$$P_{sd} = \frac{N_{EX}}{N_{TO}}, \tag{27}$$

where N_{TO} is the total number of Monte Carlo trials; N_{EX} denotes the number of event $\{\gamma(\boldsymbol{s}) \geq \gamma_0\}$ occurred during the Monte Carlo trials. The P_{sd} curves versus σ^2 for Target 1 and Target 2 are illustrated in Fig 10. One can see from the figure that the two metrics achieve almost the same performance in terms of the P_{sd} in low SINR situations. By contrast, the proposed PRD metric tends to get a higher P_{sd} value in high SINR situations, showing strong probabilistic robustness.

VI. Conclusion

This paper considers radar waveform design problem for extended targets under PAR constraint, where a novel waveform design metric with probabilistically robust properties is proposed. The novel PRD waveform design metric is derived based on the stochastic model of TIR. The geometric interpretation and limit interpretation for the proposed metric are illustrated for better understanding. Since the proposed waveform design model results in a non-convex optimization problem, we study its SDR version and devise algorithms to get high qualified (sub-optimal) waveforms. The convergences of the proposed algorithms are proved theoretically.

At the analysis stage, experiments are carried out to test the PRD waveform design metric. Results show that the proposed metric behaves very similarly to the current design metric in low SINR situations, but outperforms its counterpart in high SINR situations, exhibiting strong probabilistically robust abilities.

Our future researches might consider the extension of the proposed metric to signal dependent clutter environments as well as the joint design of waveform and filter. In addition, PRD waveform design under some practical waveform constraints, including similarity and spectral compatibility, is also the left work.

Appendix A
Proof of Proposition 1

Define $\boldsymbol{v} = \boldsymbol{R}_t^{-1/2}\boldsymbol{t}$, then we have

$$\gamma(\boldsymbol{s}) = \boldsymbol{v}^{\mathrm{H}}\boldsymbol{R}_t^{1/2}\boldsymbol{S}^{\mathrm{H}}\boldsymbol{R}_n^{-1}\boldsymbol{S}\boldsymbol{R}_t^{1/2}\boldsymbol{v}. \tag{28}$$

Further, denote by

$$\boldsymbol{R}_t^{1/2}\boldsymbol{S}^{\mathrm{H}}\boldsymbol{R}_n^{-1}\boldsymbol{S}\boldsymbol{R}_t^{1/2} = \boldsymbol{U}^{\mathrm{H}}\boldsymbol{\Sigma}\boldsymbol{U} \tag{29}$$

the eigenvalue decomposition of $\boldsymbol{R}_t^{1/2}\boldsymbol{S}^{\mathrm{H}}\boldsymbol{R}_n^{-1}\boldsymbol{S}\boldsymbol{R}_t^{1/2}$, then we recast (28) as

$$\gamma(\boldsymbol{s}) = \boldsymbol{v}^{\mathrm{H}}\boldsymbol{U}^{\mathrm{H}}\boldsymbol{\Sigma}\boldsymbol{U}\boldsymbol{v}. \tag{30}$$

Let $\tilde{\boldsymbol{v}} = \boldsymbol{U}\boldsymbol{v}$, then we get $\gamma(\boldsymbol{s}) = \sum_{q=1}^{Q}\lambda_q|\tilde{\boldsymbol{v}}(q)|^2$, where λ_q is the qth eigenvalue of $\boldsymbol{R}_t^{1/2}\boldsymbol{S}^{\mathrm{H}}\boldsymbol{R}_n^{-1}\boldsymbol{S}\boldsymbol{R}_t^{1/2}$. Note that $\boldsymbol{v} \sim \mathcal{CN}(\boldsymbol{R}_t^{-1/2}\boldsymbol{t}_0, \mathbf{I}_Q)$ and $\boldsymbol{U}$ is the unitary matrix. It is easy to verify that $\tilde{\boldsymbol{v}} \sim \mathcal{CN}(\boldsymbol{U}\boldsymbol{R}_t^{-1/2}\boldsymbol{t}_0, \mathbf{I}_Q)$, i.e., $\tilde{\boldsymbol{v}}(q)$ and $\tilde{\boldsymbol{v}}(q')$ are independent for any $q \neq q'$. Consequently, we have

$$2|\tilde{\boldsymbol{v}}(q)|^2 \sim \chi_2^2\left(2\left|\tilde{\boldsymbol{t}}_0(q)\right|^2\right), \tag{31}$$

where $\chi_2^2(2|\tilde{\boldsymbol{t}}_0(q)|^2)$ denotes the non-central χ^2 distribution [31] with freedom 2 and non-central parameter $2|\tilde{\boldsymbol{t}}_0(q)|^2$; $\tilde{\boldsymbol{t}}_0 = \boldsymbol{U}\boldsymbol{R}_t^{-1/2}\boldsymbol{t}_0$. Besides, $2|\tilde{\boldsymbol{t}}_0(q)|^2$ and $2|\tilde{\boldsymbol{t}}_0(q')|^2$ are independent for any $q \neq q'$, since $\tilde{\boldsymbol{v}}(q)$ and $\tilde{\boldsymbol{v}}(q')$ are independent.

Summarizing the analyses above, we have

$$\gamma(\boldsymbol{s}) = \frac{1}{2}\sum_{q=1}^{Q}\lambda_q\chi_2^2\left(2\left|\tilde{\boldsymbol{t}}_0(q)\right|^2\right). \tag{32}$$

According to the fundamentals of χ^2 distribution, we immediately get the expectation and variance of $\gamma(\boldsymbol{s})$ with

$$\mathbb{E}\left[\gamma(\boldsymbol{s})\right] = \mathrm{tr}\left(\boldsymbol{R}_t\boldsymbol{S}^{\mathrm{H}}\boldsymbol{R}_n^{-1}\boldsymbol{S}\right) + \boldsymbol{t}_0^{\mathrm{H}}\boldsymbol{S}^{\mathrm{H}}\boldsymbol{R}_n^{-1}\boldsymbol{S}\boldsymbol{t}_0$$

and

$$\mathbb{V}\left[\gamma(\boldsymbol{s})\right] = \mathrm{tr}\left(\boldsymbol{R}_t\boldsymbol{S}^{\mathrm{H}}\boldsymbol{R}_n^{-1}\boldsymbol{S}\boldsymbol{R}_t\boldsymbol{S}^{\mathrm{H}}\boldsymbol{R}_n^{-1}\boldsymbol{S}\right) + \boldsymbol{t}_0^{\mathrm{H}}\boldsymbol{S}^{\mathrm{H}}\boldsymbol{R}_n^{-1}\boldsymbol{S}\boldsymbol{R}_t\boldsymbol{S}^{\mathrm{H}}\boldsymbol{R}_n^{-1}\boldsymbol{S}\boldsymbol{t}_0.$$

Hence, we have completed the proof of *Proposition 1*.

Appendix B
Proof of Proposition 2

The proof of *Proposition 2* is divided into two steps.

First, we prove that $\tilde{\mathcal{P}}_{\mathrm{h}}$ shares the same optimal solution with $\tilde{\mathcal{P}}$ under high SINR situations. Denote by

$$\tilde{\mathrm{f}}(\boldsymbol{R}_s) = \frac{\mathrm{tr}(\boldsymbol{R}_t\mathbf{M}) + \boldsymbol{t}_0^{\mathrm{H}}\mathbf{M}\boldsymbol{t}_0 - \gamma_0}{\sqrt{\mathrm{tr}(\boldsymbol{R}_t\mathbf{M}\boldsymbol{R}_t\mathbf{M}) + \boldsymbol{t}_0^{\mathrm{H}}\mathbf{M}\boldsymbol{R}_t\mathbf{M}\boldsymbol{t}_0}},$$

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and $\hat{\boldsymbol{R}}_s$ the optimal solution of $\tilde{\mathcal{P}}$. Then, we have

$$\tilde{\mathrm{f}}\left(\hat{\boldsymbol{R}}_s\right) \geq \tilde{\mathrm{f}}(\boldsymbol{R}_{s_0}) \geq 0. \tag{33}$$

The left inequality holds for the reason that $\hat{\boldsymbol{R}}_s$ solves $\tilde{\mathcal{P}}$. While the right inequality holds due to the high SNR condition (13).

As an immediate result of (33), we get

$$\mathrm{tr}\left(\boldsymbol{R}_t\hat{\mathbf{M}}\right) + \boldsymbol{t}_0^{\mathrm{H}}\hat{\mathbf{M}}\boldsymbol{t}_0 - \gamma_0 \geq 0, \tag{34}$$

where $\hat{\mathbf{M}} = \mathbf{M}(\hat{\boldsymbol{R}}_s)$. Namely, $\hat{\boldsymbol{R}}_s$ also meets the constraints in $\tilde{\mathcal{P}}_{\mathrm{h}}$ and solves $\tilde{\mathcal{P}}_{\mathrm{h}}$.

Second, we prove that $\tilde{\mathrm{f}}(\boldsymbol{R}_s)$ is quasi-convex with respect to $\boldsymbol{R}_s$. It is equivalent to proving that $\tilde{\mathrm{f}}(\boldsymbol{R}_s)$ is quasi-convex with respect to $\mathbf{M}$, since $\mathbf{M}$ is linear with respect to $\boldsymbol{R}_s$.

The following analyses show that $\sqrt{\mathrm{tr}(\boldsymbol{R}_t\mathbf{M}\boldsymbol{R}_t\mathbf{M}) + \boldsymbol{t}_0^{\mathrm{H}}\mathbf{M}\boldsymbol{R}_t\mathbf{M}\boldsymbol{t}_0}$ is convex with respect to $\mathbf{M}$.

Note that both $\boldsymbol{R}_t$ and $\mathbf{M}$ are Hermitian semi-definite matrices, then we have

$$\mathrm{tr}(\boldsymbol{R}_t\mathbf{M}\boldsymbol{R}_t\mathbf{M}) = \mathrm{tr}\left(\boldsymbol{R}_t^{1/2}\mathbf{M}\boldsymbol{R}_t^{1/2}\left(\boldsymbol{R}_t^{1/2}\mathbf{M}\boldsymbol{R}_t^{1/2}\right)^{\mathrm{H}}\right)$$

and

$$\boldsymbol{t}_0^{\mathrm{H}}\mathbf{M}\boldsymbol{R}_t\mathbf{M}\boldsymbol{t}_0 = \left(\boldsymbol{R}_t^{1/2}\mathbf{M}\boldsymbol{t}_0\right)^{\mathrm{H}}\boldsymbol{R}_t^{1/2}\mathbf{M}\boldsymbol{t}_0.$$

To this end, we get

$$\begin{aligned}
&\sqrt{\mathrm{tr}(\boldsymbol{R}_t\mathbf{M}\boldsymbol{R}_t\mathbf{M}) + \boldsymbol{t}_0^{\mathrm{H}}\mathbf{M}\boldsymbol{R}_t\mathbf{M}\boldsymbol{t}_0} \\
&= \sqrt{\left\|\boldsymbol{R}_t^{1/2}\mathbf{M}\boldsymbol{R}_t^{1/2}\right\|_F^2 + \left\|\boldsymbol{R}_t^{1/2}\mathbf{M}\boldsymbol{t}_0\right\|_F^2} \\
&= \left\|\boldsymbol{R}_t^{1/2}\mathbf{M}\left[\boldsymbol{R}_t^{1/2}, \boldsymbol{t}_0\right]\right\|_F,
\end{aligned} \tag{35}$$

where $\|\cdot\|_F$ is the Frobenius norm. According to the convexity of Frobenius norm, we have proved that $\sqrt{\mathrm{tr}(\boldsymbol{R}_t\mathbf{M}\boldsymbol{R}_t\mathbf{M}) + \boldsymbol{t}_0^{\mathrm{H}}\mathbf{M}\boldsymbol{R}_t\mathbf{M}\boldsymbol{t}_0}$ is convex with respect to $\mathbf{M}$.

Furthermore, $\mathrm{tr}(\boldsymbol{R}_t\mathbf{M}) + \boldsymbol{t}_0^{\mathrm{H}}\mathbf{M}\boldsymbol{t}_0 - \gamma_0$ is linear with respect to $\mathbf{M}$. The constraints set is convex and guarantees $\mathrm{tr}(\boldsymbol{R}_t\mathbf{M}) + \boldsymbol{t}_0^{\mathrm{H}}\mathbf{M}\boldsymbol{t}_0 - \gamma_0 \geq 0$.

Summarizing the analyses above, we get that $\tilde{\mathcal{P}}_{\mathrm{h}}$ is a quasi-convex optimization problem with respect to $\mathbf{M}$.

Therefore, we have completed the proof of *Proposition 2*.

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